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Strength recovery after anterior cruciate ligament reconstruction with quadriceps tendon versus hamstring tendon autografts in soccer players: A randomized controlled trial☆

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ABSTRACT

Background: The comparison between HT and QT grafts in strength recovery and function after an ACLR is scarce in the literature.
Methods: A total of 56 participants were enrolled in this randomized controlled trial and placed into two groups: HT or QT. The hamstring/quadriceps (H/Q) ratio was the primary end-point measured with a Genu-3 dynamometer. Peak torque, functional assessment (Lysholm knee scoring scale and Cincinnati Knee Rating System), and anteroposterior laxity (KT-2000™ arthrometer) were also assessed. An intention-to-treat analysis was performed.
Results: The results of the H/Q ratio analysis of the participants over time revealed significant differences at 60, 180, and 300°/s at three, six, and 12 months of follow-up (60°/s: $F = 5.3$, $p = 0.005$; 180°/s: $F = 5.5$, $p = 0.004$; 300°/s: $F = 5.1$, $p = 0.005$). Furthermore, they revealed significant differences at 60°/s, 180°/s, and 300°/s in the participants over time for peak torque in the extensor muscle strength at three and six months of follow-up, with higher values in the hamstring tendon group but not at 12 months of follow-up. There were no significant differences in functional endpoints or arthrometer assessments at 24 months of follow-up.
Conclusion: An ACLR with a QT graft showed similar functional results with a better isokinetic H/Q ratio compared to an ACLR with the HT at 12 months of follow-up in soccer players. This higher H/Q ratio observed with the QT could be an advantage of this graft over the HT for an ACLR.

☆ Clinical trial registration: Clinical Trials.gov NCT0283279: https://clinicaltrials.gov/ct2/show/NCT02832791?term=cantarero+villanueva&rank=3.

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1. Introduction

The anterior cruciate ligament (ACL) is the most commonly injured ligament of the knee. Playing soccer increases the risk of suffering this injury [1]. An ACL-deficient knee can lead to recurrent instability, meniscus tears, and osteoarthritis [2]. The ACL reconstruction (ACLR) is the standard injury treatment for those soccer players who wish to return to play [3]. The choice of the graft for an ACLR remains controversial; allograft or autografts can be used, but the use of an allograft should be avoided in soccer players because of the high re-rupture rate observed in young, active patients [4]. Many authors consider a bone–patellar tendon (BPT) autograft to be the gold-standard graft for an ACLR in contact athletes due to its biomechanical properties [5]. Concerns about the BPT autograft's donor site morbidity have led many surgeons to use a hamstring tendon (HT) autograft for an ACLR [6,7]. Nevertheless, HT autografts are associated with a flexion strength deficit in isokinetic testing 8–13.

A quadriceps tendon (QT) graft could be an interesting alternative for an ACLR in soccer players. Shani et al. [14] showed that the structural properties of the QT are significantly higher than those of the BPT. Previous studies have compared the QT and the BPT, showing a reduced incidence of donor site morbidity when the QT was used [15,16]. Besides, maintaining the hamstrings' integrity after an ACL injury could help to protect the knee in valgus pivoting sport activities [17,18]. Nevertheless, no randomized controlled trial has compared QT and HT grafts for an ACLR.

Strength recovery is one of the most important parameters considered to return to play after an ACLR [19]. This strength recovery might be even more important in the case of soccer players returning to their pre-injury activity level. Analyses of strength recovery after an ACLR with the QT are scarce in the literature. To date, the quadriceps/hamstrings (Q/H) ratio, which is considered a relevant muscle balance parameter, has not been reported in a comparative study involving QT autografts and other graft choices for an ACLR. Most surgeons allow patients to return to play without restrictions between six and nine months after an ACLR [20]. The flexion and extension strength recovery and the H/Q ratio after an ACLR with the QT could be important parameters for choosing the QT among other available grafts.

The purpose of this study was to prospectively compare the strength recovery and functional outcomes of an anatomic single bundle reconstruction with HT and QT autografts in competitive soccer players. The main hypothesis of this study was that an ACLR with the HT and ACLR with the QT yield similar knee stability, functional results, and extension muscle strength recovery with a significant deficit in the isokinetic hamstring muscle strength in patients who had an ACLR with the HT compared to patients who had an ACLR with the QT.

2. Methods

2.1. Design

This was a prospective randomized controlled trial (Clinical Trials.gov NCT0283279: https://clinicaltrials.gov/ct2/show/NCT02832791?term=cantarero+villanueva&rank=3). The trial was approved by the relevant Ethical Organization and was performed according to the Helsinki Declaration (last modification in 2000) and The Biomedical Research (14/207). All participants provided a written informed consent, which was signed by a legal representative in the case of minor participants.

2.2. Participants

Fifty-six consecutive participants were enrolled by their surgeon from the Andalusian Mutuality of soccer players (Spain) for a year from July 2015. The inclusion criteria were i) having suffered an ACL injury with less than six months of evolution of the lesion at the time of diagnosis and ii) being competitive soccer players. The patients were excluded i) if they had had a previous joint injury or surgery on the affected knee, ii) if they had had any concomitant ligament injury, iii) if they had been removed more than 50% of either the lateral or medial meniscus, and iv) if they had had any articular cartilage lesion greater than Outerbridge grades I-II.

2.3. Randomization

The participants were randomized using two randomized number cycles with a computer. This sequence was introduced in numbered, opaque, sealed envelopes by an external researcher who was not a participant in the study. Once the participants were evaluated and were assigned an identification number, the surgeon checked their group.

2.4. Rehabilitation

The rehabilitation protocol was based on a previous study by Grinsven [21]. Therefore, both groups followed the same staged protocol (Table 1), except those patients with a concomitant meniscal repair, who had a restriction over 90° of flexion for four weeks. The rehabilitation was performed by the same team of physical therapists in both groups.

Table 1
Rehabilitation protocol for an ACL reconstruction.

Phases	Objectives			
Pre-surgery	Decrease of pain and inflammation Full achievement of the range of motion of the knee (ROM) Prevent muscle atrophy and normalize gait pattern			
Phases	Objectives			
	Phase 1 (week 1)	Phase 2 (2–9 weeks)	Phase 3 (10–16 weeks)	Phase 4 (17–24 weeks)
Post-surgery	Controlling pain and inflammation ROM: 0°–90° Patella mobilizations Regaining muscle control: open chain (90°–40°), closed chain (0°–60°) Improving gait pattern: if sufficient neuromuscular control, aim walking without crutches from day 4	ROM: 0–120° week 3: 0–130° week 5 Increasing ROM for open and closed chain exercises. Start static cycling: week 4 Stair stepping machine: week 4 Outdoor cycling: week 8 Jogging in straight line: week 8	Obtaining and maintaining full ROM Optimizing muscle strength and endurance Neuromuscular training with dynamic stability and plyometric exercises	Maximizing muscle strength, endurance, and neuromuscular control. Turning and cutting maneuvers are allowed
Phases	Objectives			
	Criteria to start with phase 2	Criteria to start with phase 3	Criteria to start with phase 4	Criteria for returning to sports
Post-surgery	Painless ROM 0°–90° Minimal swelling Good patellar mobility Sufficient quadriceps control to perform a mini-squat and straight leg raises in all directions	Minimal pain and swelling Painless ROM: 0–130° Normal gait pattern	No pain or swelling Full ROM Hop tests 75% > compared to the contralateral side	Hop test 85 > compared to contralateral Knee Quadriceps. Hamstring strength 85% > compared to the contralateral side.

2.5. Surgical technique

The ACL reconstructions were performed by the same surgeon. The procedure began with an arthroscopic examination through an anterolateral viewing portal to corroborate the ACL injury and determine possible associated lesions. The accessory medial portal was established slightly above the joint line, at approximately two centimeters from the medial edge of the patellar tendon. This portal was used as an instrumentation portal. A third portal, a high anteromedial portal, was made above the second portal; this portal allowed the visualization of the femoral footprint. After addressing the associated meniscal or cartilaginous lesions, the graft was harvested.

2.5.1. Quadriceps tendon graft

A four-centimeter vertical incision was made starting at the proximal pole of the patella and directed proximally and was centered in line with the quadriceps tendon. The subcutaneous tissue was dissected, and a 70-to 80-mm-long, 10-mm-wide, and seven-millimeter-deep graft was obtained with the use of a number 21 scalpel [22]. Two centimeters of both limbs of the graft were sutured with a no. 2 absorbable suture.

2.5.2. Hamstring graft

A four-centimeter oblique incision was made starting two centimeters medial to the tibial tubercle and directed proximally and medially. After dissecting the subcutaneous tissue, the sartorius fascia was incised, and both tendons were identified and harvested with the use of a tendon stripper. The hamstring graft size was less than seven millimeters of diameter; in such cases, the semitendinosus tendon was triplicated and a five-strand graft was used.

2.5.3. Preparation of the tunnels and graft fixation

The tibial tunnel was established with the use of a drill tibial guide (pinn ACL guide, CONMED, Largo, FL, USA) placed in the center of the anatomical ACL footprint, medial to the anterior horn of the lateral meniscus, and lateral to the medial tibial spine.

The tibial tunnel was drilled with a previously selected cannulated drill diameter based on the size of the graft. Then, attention was turned to the femoral tunnel. A Kirschner wire (K-wire) was placed in the center of the anatomical femoral footprint, below the lateral intercondylar ridge, and located in the center of the bifurcated ridge, which separates the insertion of the Anteromedial (AM) and Posterolateral (PL) bundles of the ACL [23]. Once the pin was in the desired position, the K-wire was drilled with the knee held in 120° of flexion. Subsequently, the tunnel was drilled with the drill bit.

The graft passage was completed with the insertion of 20 mm of the graft inside the femoral tunnel. A suspensory extracortical fixation button (RIGIDLOOP, DePuy Synthes, Raynham, MA, USA) was used in the hamstring group, and a bioabsorbable screw, two millimeters smaller than the tunnel diameter, was used in the quadriceps group. Finally, the tibial side was fixed at 20° of flexion with a bioabsorbable interference screw of the same size of the drill bit.

2.6. Outcome measurements

The participants were asked to attend a face-to-face assessment. The evaluation of endpoints was performed before the surgery and at three, six, and 12 months for all variables except for functional questionnaires and re-injury rates, which were evaluated at three, six, 12, and 24 months of follow-up. All the measurements were assessed by an expert researcher with over 10 years of experience.

2.6.1. Primary endpoint

The H/Q ratio was the primary endpoint and was measured using a Genu 3 (Easytech, Firenze, Italy) dynamometer. Previously, the patients performed 10 min of warm-up with a cycloergometer and mobility and stretching exercises. The H/Q ratio was evaluated with a protocol involving three series at 60°/s (three repetitions), 180°/s (five repetitions), and 300°/s (15 repetitions), with a 60-second recovery period between the series. The patients were seated with 90° of flexion, and the lateral femoral condyle was aligned with the rotational axis. The position was maintained with belts. The patients received verbal encouragement to maximize performance.

2.6.2. Secondary endpoints

Peak torque was calculated following the previously described protocol at 60°/s, 180°/s, and 300°/s in Newton-meters (Nm). Functional assessment was evaluated using the Lysholm knee score and Cincinnati Knee Rating System. These subjective scales allow the determination of whether the patients have returned to their level of activity prior to surgery [24], and whether they have showed a high interclass correlation coefficient (ICC) [25]. Anteroposterior laxity was measured in millimeters (mm) using the KT-2000™ arthrometer (MEDmetric, San Diego, CA, USA) with the patients placed in a supine position with a pillow under the knee to stabilize the knee at 25° of flexion. The arthrometer was placed in line with the tibia using Velcro straps. The arthrometer KT-2000™ was found to have reliable testing with manual maximum forces [26].

Table 2
Demographic and evolution of recovery data of the groups.

Items	QT group (n = 26)	HT group (n = 25)	p value
Demographic characteristics of the patients			
Age (year), mean (SD)	18.7 ± 3.6	19.2 ± 3.6	0.61[a]
Gender, n (%)			
Male	23 (88.5)	16 (64.0)	0.05
Female	3 (11.5)	9 (36.0)	
Injured side, n (%)			
Right	12 (46.2)	13 (52.0)	0.78
Left	14 (53.8)	12 (48.0)	
Educational level, n (%)			
Primary schooling	8 (30.8)	7 (28.0)	0.19
Secondary schooling	15 (67.7)	10 (40.0)	
University	3 (11.5)	8 (32.0)	
Tobacco			
No	26 (100)	24 (96.0)	0.49
Yes	0 (−)	1 (4.0)	
Alcohol intake			
Never	15 (57.7)	16 (64.0)	0.64
Monthly	10 (38.8)	7 (28.0)	
Weekly	1 (3.8)	2 (8.0)	
Body mass index (kg/cm^2), mean (SD)	23.0 ± 2.2	23.5 ± 3.5	0.55
Time playing (years), mean (SD)	10.1 ± 3.8	10.2 ± 4.1	0.93
Evolution of recovery			
Time attending the treatment protocol (months), mean (SD)	1.2 ± 0.4	1.1 ± 0.4	0.66
Practice of crutch from the 4th day, n (%)			
No	7 (26.9)	2 (8.0)	0.79
Yes	19 (73.1)	23 (92.0)	
Practice of the bicycle from the 3rd week, n (%)			
No	11 (42.3)	6 (24.0)	0.13
Yes	15 (57.7)	19 (76.0)	
Practice of running from the 3rd month, n (%)			
No	1 (3.8)	5 (20.0)	0.08
Yes	25 (96.2)	20 (80.0)	
Practice normal training at 6 months, n (%)			
No	11 (65.4)	14 (56.0)	0.34
Yes	9 (34.6)	11 (44.0)	

[a] The values are expressed as the mean and SD or frequency n (%). p values of between group differences using t test for independent samples (continuous variables) and X^2 analysis (categorical variables).

2.7. Data analysis

IBM Statistical Package for the Social Sciences (SPSS) version 22.0 was used for the analysis, and we considered the level of significance at five percent. The normal distribution of variables was checked with the Shapiro–Wilk test. The demographic data are expressed as the mean and standard deviation (SD) or frequency n (%), and we used Student's t and Chi-square tests to examine the differences in the baseline outcome variables between groups. A repeated measures analyses of the covariance (ANCOVA) was the main analysis for (i) isokinetic strength, Lysholm knee score, and arthrometer results as dependent variables, (ii) groups (the QT and HT groups) as between-subject variables, (iii) time (preoperative and at three, six, 12, and 24 months of follow-up) as within-subject variable, and (iv) age, gender, alcohol, and tobacco as covariates. For pair-wise comparisons (post hoc), we used Bonferroni's adjustments. We used the intention-to-treat principle for all analyses, with the worst-case value for

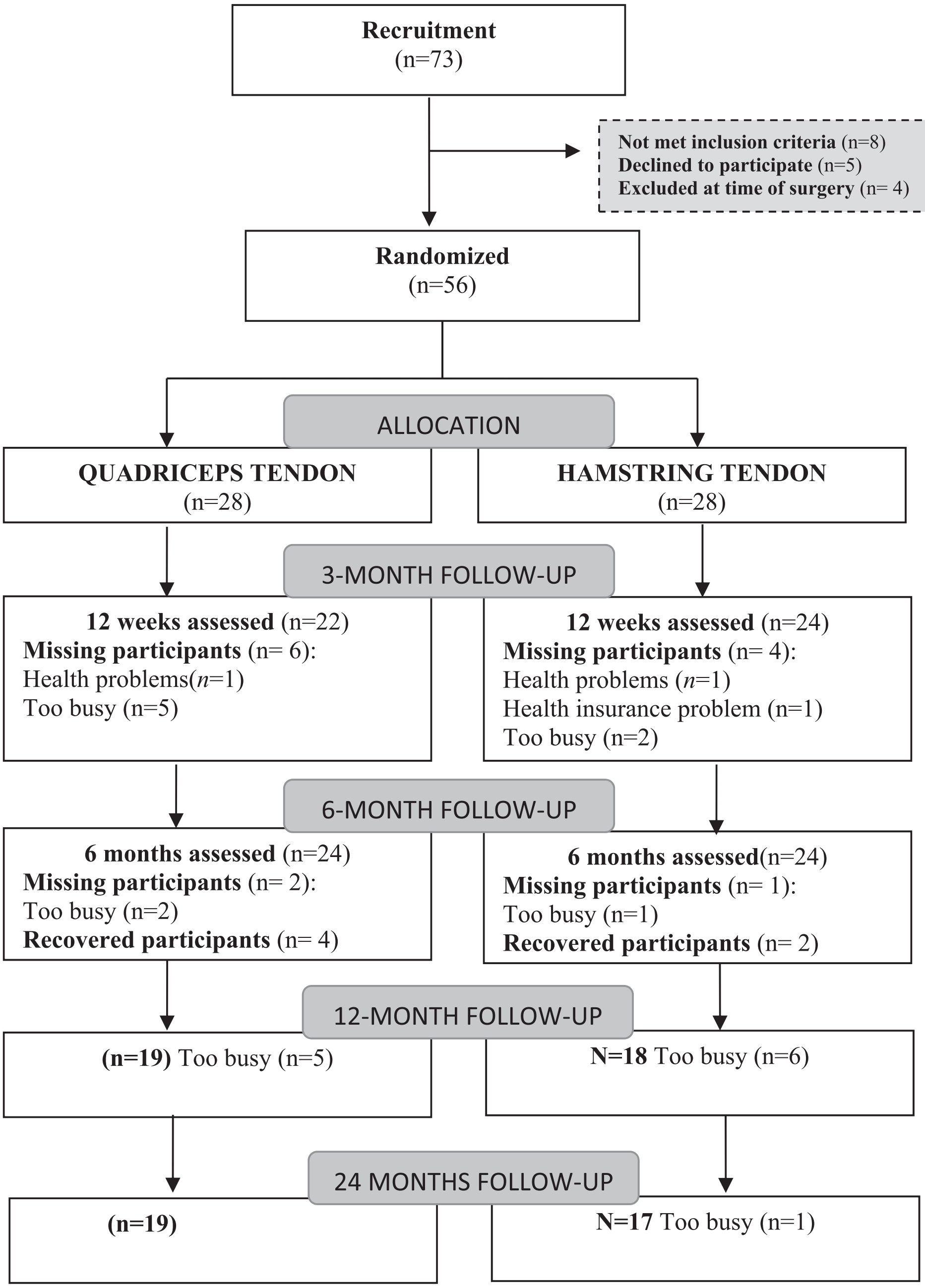

Figure 1. Study flow diagram.

missing data [27]. To assess re-rupture graft proportion incidence, a Chi-square test was used. When we only had one value of data for a patient, that value was removed from the analysis.

3. Results

Of the 73 patients, 56 were randomized into two groups of 28 patients (the QT group and the HT group), and 26 (the QT group) and 25 (the HT group) were included in the statistical analysis. Seventeen patients refused to participate in the study or had at least one of the exclusion criteria before or at the time of the surgery. Table 2 shows the clinical characteristics of the sample. There were two drop-outs (7.1%) in the QT group (reasons: too busy to attend the assessments, n = 2) and three drop-outs (10.7%) in the HT group (reasons: health insurance problem, n = 1; health problems, n = 1; too busy, n = 1). Some patients could only complete two of three assessments. The patients flow is detailed in Figure 1. The rehabilitation evolution

Table 3
H/Q ratio measurements at the baseline, at 3, 6, and 12 months of the follow-up.

Angular velocity 60 deg/s			
	QT group	HT group	**Between-Group Differences**
Pre-intervention	97.9±17.3	99.6±28.4	
3 months of follow-up	153.3±57.7	99.4±25.6	
6 months of follow-up	110.3±15.4	93.2±14.1	
12 months of follow-up	120.3±39.9	97.6±16.1	
	Within Group Change		
Pre-3 months of follow-up	-55.43 (-84.9; -25.9)	0.28 (-30.9-31.4)	55.7* (24.5-86.5)
Pre-6 months of follow-up	-12.41 (-28.3-3.5)	6.47 (-10.3-23.3)	18.8* (2.1-35.7)
Pre-12 months of follow-up	-22.36 (-46.2-1.5)	2.05 (-23.2-27.3)	24.41* (0.8-49.6)
Angular velocity 180 deg/s			
	QT group	HT group	**Between-Group Differences**
Pre-intervention	95.8±16.2	100.3±24.7	
3 months of follow-up	147.6±62.3	100.9±23.7	
6 months of follow-up	111.2±11.1	91.6±26.2	
12 months of follow-up	122.5±38.3	99.7±14.4	
	Within Group Change		
Pre-3 months of follow-up	-51.8 (-22.8;-80.8)	-0.6 (-31.3;30.1)	51.1* (20.5;81.8)
Pre-6 months of follow-up	-15.4 (-32.0;1.2)	8.7 (-8.9;26.3)	24.1* (6.5;41.7)
Pre-12 months of follow-up	-26.7 (-48.8;-4.5)	0.6 (-22.8;23.9)	27.3* (3.9;50.6)
Angular velocity 300 deg/s			
	QT group	HT group	**Between-Group Differences**
Pre-intervention	104.3±18.8	109.9±26.6	
3 months of follow-up	140.8,±34.5	105.0±22.7	
6 months of follow-up	109.1±18.4	98.5±11.1	
12 months of follow-up	120.9±28.8	103.7±10.7	
	Within Group Change		
Pre-3 months of follow-up	-36.4 (-57.9;-15.0)	4.8 (17.5;27.5)	41.3* (18.7;63.9)
Pre-6 months of follow-up	-4.8 (-7.9;17.5)	11.3 (-2.1;24.8)	16.2* (2.7;29.6)
Pre-12 months of follow-up	-16.5 (-37.4;4.8)	6.2 (-15.9;28.3)	22.7* (0.6;44.7)

The values are expressed as mean and standard deviation for pre intervention and 3, 6, and 12 months of follow-up data, and as mean (95% CI) for within and between groups change score. * Significant group–time interaction (repeated ANCOVA test, $p < 0.05$).

Table 4
Peak torque (Nm) measurements at the baseline, at 3, and at 6 months of the follow-up.

Angular velocity 60 deg/s						
	Flexion			Extension		
	QT group	HT group	**Between-Group Differences**	QT group	HT group	**Between-Group Differences**
Pre-intervention	103.7±30.4	90.6±26.1		110.4±36.4	104.3±40.1	
3 months of follow-up	112.3±30.9	104.3±36.1		85.2±40.1	118.4±45.3	
6 months of follow-up	136.4±39.8	116.5±41.3		125.7±40.3	139.9±47.9	
12 months of follow-up	147.4±37.8	131.3±40.4		139.5±47.3	144.94±51.4	
	Within Group Change			**Within Group Change**		
Pre-3 months of follow-up	-8.6 (-27.9;10:5)	-13.6 (-34.0;6.7)	-7.2 (-24.8;10.3)	25.2 (0.5;49.9)	-14.1 (-40.2; 11.9)	-39.3* (-65.3;.13.4)
Pre-6 months of follow-up	-32.7 (-57.5;-7.9)	-25.8 (-52.1;0.3)	-6.6 (-9.9;27.3)	-15.4 (-42.8; 12.1)	-35.6 (-64.7; -6.5)	19.0* (-3.3;41.4)
Pre-12 months of follow-up	-43.7 (-67.4;-20.1)	-40.7 (-65.7;-15.7)	-2.8 (-16.5;19.3)	-29.2 (-60.0;1.7)	-40.6 (-73.8;-8.0)	8.8 (-30.4;12.8)
Angular velocity 180 deg/s						
	Flexion			Extension		
	QT group	HT group	**Between-Group Differences**	QT group	HT group	**Between-Group Differences**
Pre-intervention	90.7±17.5	84.3±21.1		93.4±24.1	88.5±27.5	
3 months of follow-up	96.7±24.5	95.9±28.1		71.7±27.6	98.6±32.5	
6 months of follow-up	117.9±30.7	109.7±34.2		104.0±29.1	113.6±32.3	
12 months of follow-up	127.4±34.1	112.2±29.3		115.6±33.2	114.7±32.9	
	Within Group Change			**Within Group Change**		
Pre-3 months of follow-up	-6.0 (-21.1;9.1)	-11.5 (-27.5;4.4)	-5.5 (-21.5;10.4)	21.6 (7.2;36.1)	-10.1 (-25.4;5.1)	-31.8* (-47.0; 16.5)
Pre-6 months of follow-up	-27.1 (-46.9-7.4)	-25.4 (-46.3;-4.4)	7.2 (-9.3;23.9)	-10.6 (-28.1;6.8)	-25.0 (-43.5;-6.5)	-17.3* (-32.1;-2.7)
Pre-12 months of follow-up	-36.7 (-57.1;-16.3)	-27.8 (-49.4;-6.2)	7.1 (-21.0;6.8)	-22.2 (-42.4;-1.9)	-26.2 (-47.5;-4.8)	10.4 (-28.7;7.8)
Angular velocity 300 deg/s						
	Flexion			Extension		
	QT group	HT group	**Between-Group Differences**	QT group	HT group	**Between-Group Differences**
Pre-intervention	73.2±10.9	71.1±16.2		69.0±15.6	67.4±23.3	
3 months of follow-up	78.7±18.0	78.7±18.1		58.7±19.3	75.8±24.6	
6 months of follow-up	88.4±19.8	82.3±18.3		81.5±19.7	84.6±22.4	
12 months of follow-up	99.0±18.1	84.8±17.2		87.0±21.3	82.0±19.9	
	Within Group Change			**Within Group Change**		
Pre-3 months of follow-up	-5.4 (-14.9;3.9)	-4.1 (-14.1;5.9)	1.4 (-8.6;11.4)	10.2 (-2.1;22.6)	-8.4 (-21.4;4.6)	-18.6* (-31.7;-5.6)
Pre-6 months of follow-up	15.1 (-27.5;-2.7)	-11,2 (-24.3;1.9)	2.6 (-8.9; 14.3)	-12.5 (-25.4;0.3)	-17.2 (-30.8;-3.6)	13.9* (0.8;27.1)
Pre-12 months of follow-up	-27.7 (-36.5;-14.9)	-13.7 (-25.1;-2.3)	8.1 (-2.9;19.0)	-18.0 (-33.1;-2.8)	-14.6 (-30.6;1.4)	8.1 (-20.5;4.3)

The values are expressed as mean and standard deviation for pre intervention and 3 and 6 months of follow-up data and as mean (95% CI) for within and between groups change score. * Significant group–time interaction (repeated ANCOVA test, $p < 0.05$).

was similar in both groups (Table 2). The average time in the treatment protocol was 1.2 ± 0.4 months in the QT group and 1.1 ± 0.4 months in the HT group. One patient did not want to do the isokinetic assessment at three months because he did not feel like he could do so safely.

The results of the analysis of the H/Q ratio of the participants over time revealed significant differences at 60°/s, 180°/s, and 300°/s at three, six, and 12 months of follow-up (60°/s: $F = 6.5$, $p = 0.003$; 180°/s: $F = 4.8$, $p = 0.012$; 300°/s: $F = 5.97$, $p = 0.005$), with higher values in the QT group (Table 3). The inclusion of covariates did not modify the results.

The repeated ANCOVA of the participants over time revealed significant differences at 60°/s, 180°/s, and 300°/s ($F = 4.70$, $p = 0.014$; $F = 6.23$, $p = 0.004$; $F = 2.91$, $p = 0.04$) for peak torque in extensor muscle strength at three, six, and 12 months of follow-up. The HT group showed a greater increase in peak torque in extension than the QT group did at the three velocities (Table 4). The post hoc analysis did not display significant differences between groups at 12 months of follow-up. The inclusion of

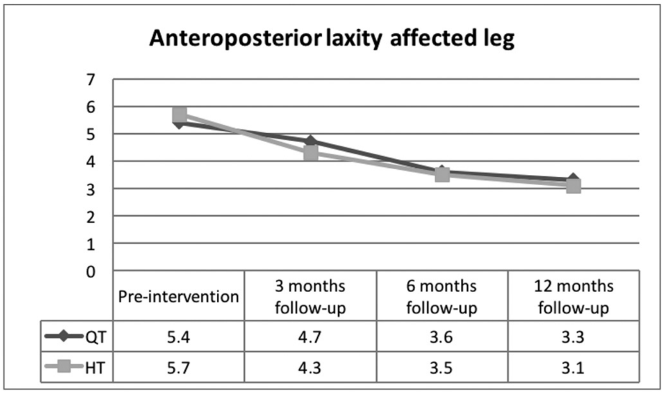

Figure 2. Anteroposterior laxity values assessment with arthrometer in the injured knee.

covariates did not modify the results. The flexion strength analysis of the participants over time showed non-significant results for any velocity.

The results of the repeated ANCOVA did not show any significant differences ($p = 0.719$ injured knee; $p = 0.993$ no injured knee) between the QT and HT groups in arthrometry with different moments of evaluation (Figure 2). There was no significant difference between the groups in postoperative assessment in functional endpoints evaluated using the Lysholm knee scoring ($p = 0.892$) and the Cincinnati Knee Rating System ($p = 0.512$). The values gradually recovered from the baseline to 24 months of follow-up (Figure 3).

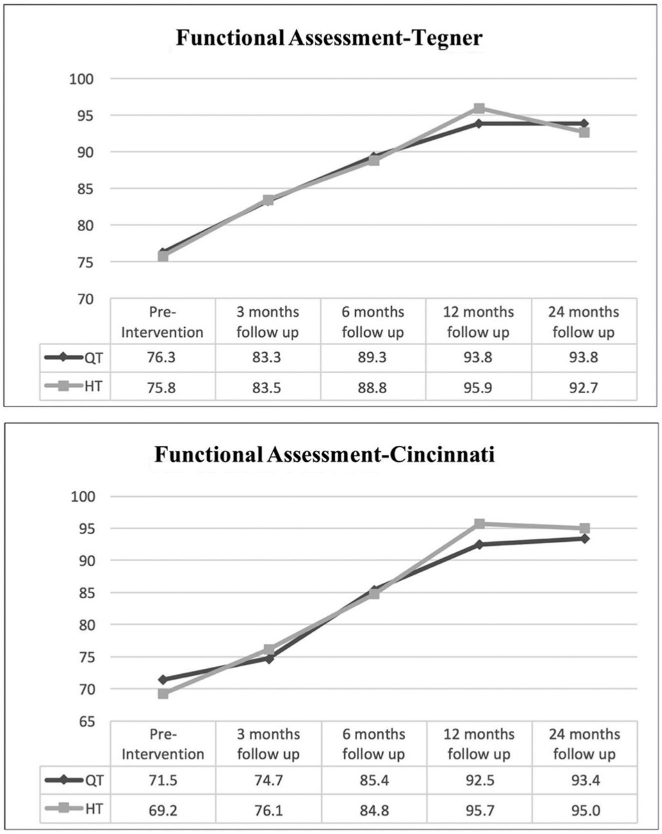

Figure 3. a) Functional assessment values obtained with the Tegner activity level scale. b) Functional assessment values obtained with the Cincinnati Knee Rating System.

During the follow-up there were three graft ruptures in the HT group and one graft rupture in the QT group. All these re-ruptures occurred playing soccer between the seventh and eleventh months after the surgery. A tendency to a significant increase in rupture graft was found in the HT group ($X^2 = 3.31$; $p = 0.10$).

4. Discussion

This study is a randomized controlled trial comparing an ACLR with the QT without bone block and an ACLR with the HT in competitive soccer players. It was performed to compare the knee stability and strength recovery 12 months after an ACLR with both grafts, and the functional results after 24 months of follow-up.

Soccer is the most popular sport worldwide. Playing soccer is a risk for an ACL injury, and soccer players are the population with the highest odds for a revision surgery after an ACLR [28,29]. Therefore, the graft used in an ACLR in soccer players should have suitable biomechanical properties for a low revision rate and low morbidity. The most popular grafts used for an ACLR are the BPT and the HT. Both have advantages and disadvantages. The BPT has excellent biomechanical properties with a lower rate of revision than the HT [5]; however, because its donor site morbidity and extension strength deficit [30] are a concern for many surgeons, the use of the HT has increased in the recent years [6,7]. The quadruple-strand HT graft is stiffer and has higher loads compared to a 10 mm BPT [31], but it is associated with a higher revision rate, so there might be some factors that influence the failure rate in addition to its intrinsic biomechanical properties. Previous isokinetic testing with the HT has shown similar results to this study with a flexion strength deficit 8–13. Hamstring activation reduces the ACL load providing knee stability [32,33]. Therefore, preserving the hamstrings during an ACLR could protect the graft, thereby decreasing the risk of rupture of the ACL graft. The QT was described for the first time by Marshall in 1979 [34]; since then, it has been the least used autograft [35], despite the good results shown by Fulkerson two decades ago [36,37]. However, it could have some advantages over the two previously mentioned grafts. It is a more versatile graft than the HT, the QT can be used as single or double bundle graft with or without bone block, respectively [38], and maintaining the hamstrings' integrity after an ACL injury could help to protect knee in valgus pivoting sport activities [17,18]. Recently, Shani [14] published results showing its superior biomechanical properties, with nearly twice the cross sectional area and a higher stiffness and load to failure compared to the BPT. Different authors have reported excellent results with the use of bone block autograft and the QT for an ACLR [39], with less donor-side morbidity compared to the BPT [19]. Thus, based on the current literature, the QT might be at least as good a graft option as the BPT is [40,41], although long-term follow-up studies should confirm this in a clinical setting.

Investigations comparing the HT and the QT are scarce in the literature. In a cohort study, Lee et al. [42] compared the ACLR with a double-bundle HT and quadriceps tendon with bone block autograft. They found similar results in both groups in terms of knee stability and functional outcomes and a better flexor strength recovery in the QT graft. Although the results of our study are similar to those reported by Lee, we did not find significant differences in peak torque after 12 months, although differences in the H/Q ratio were found between the groups. Lee et al. [42] used a QT with bone block, whereas in our study, we used a free central quadriceps graft (QT) without bone block. We believe that harvesting the quadriceps graft with a bone plug is not necessary because an adequate length can be obtained without it [22], and it could increase the risk of patellar fracture [39,43–45].

Although it might be intuitive that harvesting a central portion of the quadriceps tendon might affect the strength of extension [46], this was not confirmed in our study. We did not find a quadriceps strength loss during the follow-up period unlike Chen et al. [47]. The different methods used to assess the quadriceps strength could explain the differences between the results of both studies. The results of the isokinetic testing in our study showed a significantly different H/Q isokinetic ratio between the grafts, with an increased ratio in all of the tested velocities in the QT group. The reported ratios in the QT grafts group exceeded 1.0, which is specifically recommended for preventing ACL injuries [48], whereas the HT group showed a reduction close to 1.0 at 12 months of follow-up. This is the first time that this ratio has been reported in a comparative study between QT grafts and any other graft for an ACLR. This ratio imbalance is a relevant predictor of muscle injuries in soccer players [49] and might impact in-season soccer performance [50]. However, a better hamstring strength recovery observed in the QT group could be a major determinant of the H/Q ratio imbalance after an ACL reconstruction.

Harris et al. [20] reported that 51% of surgeons allow their patients a full range of activities without restrictions six months after ACLR, and that 86% permit return to play nine months after an ACLR, despite the fact that many re-ruptures occur in the first 12 months after the procedure [51]. Co-activation of the hamstrings and quadriceps might be critical to decrease the risk of an ACL injury [17]. The strength flexion deficit might increase the risk for ACL rupture in soccer players [52]; therefore, it might contribute to the higher re-rupture rate observed in the HT compared to the BPT [53], and the higher re-rupture rate observed also in the HT group compared to the QT group observed in this study. Thus, the results of the isokinetic testing and the reported H/Q ratio indicate that the QT graft might be a better choice than the HT for soccer players.

The limitations of this study must be recognized. This was a single-surgeon study, and the results might not be generalizable. The study had a short period of follow-up of 12 months, but it was well adapted to the return-to-play time according to the primary outcomes. These preliminary results warrant a new follow-up period at two and five years to observe the evolution of this ratio and possible influences in an ACL re-injury. Another potential limitation is that the fixation method of the graft was different in the QT and the HT, with the use of interference screws and extracortical fixation button, respectively. However, literature reveals that there is no influence of the method used to fix the graft in surgery outcomes [54]. Finally, the inclusion of other interesting outcomes as rotatory stability or Varus–valgus-malalignment should be taken into account in future trials. Nevertheless, this was a high-quality design with a prospective and randomized design. All patients underwent surgery by the same experienced orthopedic surgeon. All patients followed the same rehabilitation protocol, which was conducted by the same physiotherapists, and the same researchers assessed the different study variables.

5. Conclusion

The ACLR with a QT graft showed similar functional results with a better isokinetic H/Q ratio compared to the ACLR with the HT graft at 12 months and similar functional outcome at 24 months of follow-up in soccer players. This higher H/Q ratio observed with the QT could be an advantage of this graft over the HT for an ACLR in soccer players.

List of abbreviations

ANCOVA	Analyses of the covariance
ACL	Anterior cruciate ligament
ACLR	Anterior cruciate ligament reconstruction
BPT	Bone–patella tendon
HT	Hamstring tendon
H/Q	Hamstring/quadriceps
mm	Millimeters
Nm	Newton-meters
QT	Quadriceps tendon
SD	Standard deviation

Competing interests

None.

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References

[1] Dragoo JL, Braun HJ, Durham JL, Chen MR, Harris AH. Incidence and risk factors for injuries to the anterior cruciate ligament in National Collegiate Athletic Association football: data from the 2004–2005 through 2008–2009 National Collegiate Athletic Association Injury Surveillance System. Am J Sports Med 2012;40(5):990.
[2] Roos H, Adalberth T, Dahlberg L, Lohmander LS. Osteoarthritis of the knee after injury to the anterior cruciate ligament or meniscus: the influence of time and age. Osteoarthritis Cartilage 1995;3:261–7.
[3] Nebelung W, Wuschech H. Thirty-five years of follow-up of anterior cruciate ligament-deficient knees in high-level athletes. Arthroscopy 2005;21:696–702.
[4] Kaeding CC, Pedroza AD, Reinke EK, Huston LJ. Risk factors and predictors of subsequent ACL injury in either knee after ACL reconstruction: prospective analysis of 2488 primary ACL reconstructions from the MOON cohort. Am J Sports Med 2015;43:1583–90.
[5] Reinhardt KR, Hetsroni I, Marx RG. Graft selection for anterior cruciate ligament reconstruction: a level I systematic review comparing failure rates and functional outcome. Orthop Clin N Am 2010;41:249–62.
[6] Feller JA, Webster KE. A randomized comparison of patellar tendon and hamstring tendon anterior cruciate ligament reconstruction. Am J Sports Med 2003;31:564–73.
[7] Waldén M, Hägglund M, Magnusson H, Ekstrand J. Anterior cruciate ligament injury in elite football: a prospective three-cohort study. Knee Surg Sports Traumatol Arthrosc 2011;19:11–9.
[8] Ardern CL, Webster KE, Taylor NF, Feller JA. Hamstring strength recovery after hamstring tendon harvest for anterior cruciate ligament reconstruction: a comparison between graft types. Arthroscopy 2010;26:462–9.
[9] Burks RT, Crim J, Fink BP, Boylan DN, Greis PE. The effects of semitendinosus and gracilis harvest in anterior cruciate ligament reconstruction. Arthroscopy 2005;21:1177–85.
[10] Keays SL, Bullock-Saxton J, Keays AC, Newcombe P. Muscle strength and function before and after anterior cruciate reconstruction. Knee 2001;8:229–34.
[11] Kim JG, Yang SJ, Lee YS, Shim JC, Ra HJ, Choi JY. The effects of hamstring harvesting on outcomes in anterior cruciate ligament-reconstructed patients: a comparative study between hamstring-harvested and -unharvested patients. Arthroscopy 2011;27:1226–34.
[12] Maletis GB, Inacio MC, Funahashi TT. Risk factors associated with revision and contralateral anterior cruciate ligament reconstructions in the Kaiser Permanente ACLR registry. Am J Sports Med 2015;43:641–7.
[13] Nakamura N, Horibe S, Sasaki S, Kitaguchi T, Tagami M, Mitsuoka T, et al. Evaluation of active knee flexion and hamstring strength after anterior cruciate ligament reconstruction using hamstring tendons. Arthroscopy 2002;18:598–602.
[14] Shani RH, Umpierez E, Nasert M, Hiza EA, Xerogeanes J. Biomechanical comparison of quadriceps and patellar tendon grafts in anterior cruciate ligament reconstruction. Arthroscopy 2016;32:71–5.
[15] Geib TM, Shelton WR, Phelps RA, Clark L. Anterior cruciate ligament reconstruction using quadriceps tendon autograft. Arthroscopy 2009;25:1408–14.
[16] Lund B, Nielsen T, Faunø P, Christiansen SE, Lind M. Is quadriceps tendon a better graft choice than patellar tendon? A prospective randomized study. Arthroscopy 2014;30:593–8.
[17] Alentorn-Geli E, Myer GD, Silvers HJ, Samitier G, Romero D, Lázaro-Haro C, et al. Prevention of non-contact anterior cruciate ligament injuries in soccer players: part 1: mechanisms of injury and underlying risk factors. Knee Surg Sports Traumatol Arthrosc 2009;17:705–29.

[18] Kremen TJ, Polakof LS, Rajaee SS, Nelson TJ, Metzger MF. The effect of hamstring tendon autograft harvest on the restoration of knee stability in the setting of concurrent anterior cruciate ligament and medical collateral ligament injuries. Am J Sports Med 2018;46(1):163–70.
[19] Ellman MB, Sherman SL, Forsythe B, LaPrade RF, Cole BJ, Bach Jr BR. Return to play following anterior cruciate ligament reconstruction. J Am Acad Orthop Surg 2015;23:283–96.
[20] Harris JD, Abrams GD, Bach BR, Williams D, Heidloff D, Bush-Joseph CA, et al. Return to sport after ACL reconstruction. Orthopedics 2014;37:103–8.
[21] van Grinsven S, van Cingel RE, Holla CJ, van Loon CJ. Evidence-based rehabilitation following anterior cruciate ligament reconstruction. Knee Surg Sports Traumatol Arthrosc 2010;18:1128–44.
[22] J1 Lippe, Armstrong A, Fulkerson JP. Anatomic guidelines for harvesting a quadriceps free tendon autograft for anterior cruciate ligament reconstruction. Arthroscopy 2012;28:980–4.
[23] van Eck CF, Lesniak BP, Schreiber VM, Fu FH. Anatomic single- and double-bundle anterior cruciate ligament reconstruction flowchart. Arthroscopy 2010;26: 258–68.
[24] Padua R, Alviti F, Venosa M, Mazzola C, Padua L. The influence of graft placement on clinical outcome in anterior cruciate ligament reconstruction. Joints 2016;4 (1):12–6.
[25] Briggs KK, Lysholm J, Tegner Y, Rodkey WG, Kocher MS, Steadman JR. The reliability, validity, and responsiveness of the Lysholm score and Tegner activity scale for anterior cruciate ligament injuries of the knee: 25 years later. Am J Sports Med 2009;37:890–7.
[26] Myrer JW, Schulthies SS, Fellingham GW. Relative and absolute reliability of the KT-2000 arthrometer for uninjured knees. Testing at 67, 89, 134, and 178 N and manual maximum forces. Am J Sports Med 1996;24:104–8.
[27] Cantarero-Villanueva I, Sánchez-Jiménez A, Galiano-Castillo N, Díaz-Rodríguez L, Martín-Martín L, Arroyo-Morales M. Effectiveness of lumbopelvic exercise in colon cancer survivors: a randomized controlled clinical trial. Med Sci Sports Exerc 2016;48:1438–46.
[28] Andernord D, Björnsson H, Petzold M, Eriksson BI, Forssblad M, Karlsson J, et al. Surgical predictors of early revision surgery after anterior cruciate ligament reconstruction: results from the Swedish National Knee Ligament Register on 13,102 patients. Am J Sports Med 2014;42:1574–82.
[29] Fältström A, Hägglund M, Magnusson H, Forssblad M, Kvist J. Predictors for additional anterior cruciate ligament reconstruction: data from the Swedish national ACL register. Knee Surg Sports Traumatol Arthrosc 2016;24:885–94.
[30] Maletis GB, Cameron SL, Tengan JJ, Burchette RJ. A prospective randomized study of anterior cruciate ligament reconstruction. Am J Sports Med 2007;35:384–94.
[31] Hamner DL, Brown Jr CH, Steiner ME, Hecker AT, Hayes WC. Hamstring tendon grafts for reconstruction of the anterior cruciate ligament: biomechanical evaluation of the use of multiple strands and tensioning techniques. J Bone Joint Surg Am 1999;81:549–57.
[32] Renström P, Arms SW, Stanwyck TS, Johnson RJ, Pope MH. Strain within the anterior cruciate ligament during hamstring and quadriceps activity. Am J Sports Med 1986;14:83–7.
[33] Withrow TJ, Huston LJ, Wojtys EM, Ashton-Miller JA. Effect of varying hamstring tension on anterior cruciate ligament strain during in vitro impulsive knee flexion and compression loading. J Bone Joint Surg Am 2008;90:815–23.
[34] Marshall JL, Warren RF, Wickiewicz TL, Reider B. The anterior cruciate ligament: a technique of repair and reconstruction. Clin Orthop Relat Res 1979(143): 97–106.
[35] Budny J, Fox J, Rauh M, Fineberg M. Emerging trends in anterior cruciate ligament reconstruction. J Knee Surg 2016(Mar 28) [Epub ahead of print].
[36] Fulkerson JP, Langeland R. An alternative cruciate reconstruction graft: the central quadriceps tendon. [Technical note] Arthroscopy 1995;11:252–4.
[37] Fulkerson JP. Central quadriceps free tendon for the anterior cruciate ligament reconstruction. Oper Tech Sports Med 1999;7:195–200.
[38] Van Eck CF, Illingworth KD, Fu FH. Quadriceps tendon: the forgotten graft. [autor replay] Arthroscopy 2010;26:441–2.
[39] Lee S, Seong SC, Jo CH, Han HS, An JH, Lee MC. Anterior cruciate ligament reconstruction with use of autologous quadriceps tendon graft. J Bone Joint Surg Am 2007;89:116–26.
[40] Mulford JS, Hutchinson SE, Hang JR. Outcomes for primary anterior cruciate reconstruction with the quadriceps autograft: a systematic review. Knee Surg Sports Traumatol Arthrosc 2013;21:1882–8.
[41] Slone HS, Romine SE, Premkumar A, Xerogeanes JW. Quadriceps tendon autograft for anterior cruciate ligament reconstruction: a comprehensive review of current literature and systematic review of clinical results. Arthroscopy 2015;31:541–54.
[42] Lee JK, Lee S, Lee MC. Outcomes of anatomic anterior cruciate ligament reconstruction: bone–quadriceps tendon graft versus double-bundle hamstring tendon graft. Am J Sports Med 2016;44(9):2323–9.
[43] Lee S, Seong SC, Jo H, Park YK, Lee MC. Outcome of anterior cruciate ligament reconstruction using quadriceps tendon autograft. Arthroscopy 2004;20:795–802.
[44] Piva SR, Childs JD, Klucinec BM, Irrgang JJ, Almeida GJ, Fitzgerald GK. Patella fracture during rehabilitation after bone-patellar tendon-bone anterior cruciate ligament reconstruction: 2 case reports. J Orthop Sports Phys Ther 2009;39:278–86.
[45] Steen H, Tseng KF, Goldstein SA, Carpenter JE. Harvest of patellar tendon (bone–tendon–bone) autograft for ACL reconstruction significantly alters surface strain in the human patella. J Biomech Eng 1999;121:229–33.
[46] Chen CH, Chuang TY, Wang KC, Chen WJ, Shih CH. Arthroscopic anterior cruciate ligament reconstruction with quadriceps tendon autograft: clinical outcome in 4–7 years. Knee Surg Sports Traumatol Arthrosc 2006;14:1077–85.42.
[47] Chen CH, Chuang TY, Wang KC, Chen WJ, Shih CH. The evaluation of muscle recovery after anatomical single-bundle ACL reconstruction using a quadriceps autograft. Knee Surg Sports Traumatol Arthrosc 2016;31(3):541–54.
[48] Holcomb WR, Rubley MD, Lee HJ, Guadagnoli MA. Effect of hamstring-emphasized resistance training on hamstring:quadriceps strength ratios. J Strength Cond Res 2007;211:41–7.
[49] Dauty M, Menu P, Fouasson-Chailloux A, Ferréol S, Dubois C. Prediction of hamstring injury in professional soccer players by isokinetic measurements. Muscles Ligaments Tendons J 2016;19:116–23.
[50] Ardern CL, Pizzari T, Wollin MR, Webster KE. Hamstrings strength imbalance in professional football (soccer) players in Australia. J Strength Cond Res 2015;29: 997–1002.
[51] Pallis M, Svoboda SJ, Cameron KL, Owens BD. Survival comparison of allograft and autograft anterior cruciate ligament reconstruction at the United States Military Academy. Am J Sports Med 2012;40:1242–6.
[52] Colby S, Francisco A, Yu B, Kirkendall D, Finch M, Garrett Jr W. Electromyographic and kinematic analysis of cutting maneuvers: implications for anterior cruciate ligament injury. Am J Sports Med 2000;28:234–40.
[53] Leys T, Salmon L, Waller A, Linklater J, Pinczewski L. Clinical results and risk factors for reinjury 15 years after anterior cruciate ligament reconstruction: a prospective study of hamstring and patellar tendon grafts. Am J Sports Med 2012;40:595–605.
[54] Lubowitz JH, Schwartzberg R, Smith P. Cortical suspensory button versus interference screw fixation for knee anterior cruciate ligament soft-tissue allograft: a prospective, randomized controlled trial. Arthroscopy 2015;34(9):1733–9.